

IT 520 Homework 1 — Which Part Should You Change?

Individual work · 40 points

Your gaming PC drops frames in a game's downtown district. A vendor blames the hardware and offers three upgrades. You have two supplied records from the same 60-second run: a benchmark capture and a system-monitor report. Your job is to locate the work, use the records to find the limiting part, and recommend a change that the evidence supports.

Use the values in the **case packet included below**. Show each equation with substituted values and units; an unsupported answer earns no calculation credit. Round only if a question asks you to.

For this case, treat CPU, GPU, and SSD as three serial one-server stages. The supplied impact estimate for each quote says that the quote halves the demand of its named stage and changes no other demand. It does **not** say that the whole frame time is halved.

CASE PACKET 1002

Answer from the numbers on this page only.

Your supplied records

CAPTURE -- 60 second benchmark run, downtown district

```
frames rendered ..... 1800
average frame rate ..... 30 fps
input-to-photon latency .... 100 ms
... of which, work ..... 47 ms
... of which, queued ..... 53 ms
```

SYSTEM MONITOR -- same 60 seconds

```
CPU   busy  36000 ms   60%
GPU   busy   3600 ms    6%
SSD   busy  45000 ms   75%
video memory in use ..... 94%
```

On the table

Option	What it is	Price	Supplied impact estimate
Quote A	a faster processor	\$430	halves CPU demand
Quote B	a new graphics card	\$600	halves GPU demand
Quote C	a faster NVMe SSD	\$180	halves SSD demand
Settings change	texture streaming pool, High to Medium	\$0	halves SSD demand

Each impact estimate halves only the demand of the stage it names and changes no other demand.

1. Put the frame on the machine map (3 points)

For each action below, name the primary resource or boundary it uses: **SSD, CPU, main memory, GPU/video memory, or display I/O**. Then draw arrows to put the five actions in a defensible order.

- read an asset needed for the district;

- update the crowd simulation;
- hold the active scene data;
- render the frame and hold its image before display; and
- send the completed frame to the monitor.

2. Trace one instruction (5 points)

This is a separate stored-program model exercise, not another record from the benchmark.

```
Initial state
PC = 40           IR = empty           ALU result = empty
memory[40] = ADD 4 TO [200]
memory[41] = DISPLAY [200]
memory[200] = 28
```

Trace the instruction at address 40 through **fetch, decode, execute, and write-back**. For each phase, state what information moves or is interpreted and exactly which visible state changes. End with the final values of **PC**, **IR**, the ALU result, and **memory[200]**.

3. Check the whole-system record (4 points)

1. Recalculate throughput **X** in frames/second from frames completed and the capture duration. (1)
2. Use Little's Law, $N = X R$, to calculate the average frames in the system. Convert the reported latency to seconds before multiplying. (2)
3. The three resource demands sum to the capture's **work** time, not its full input-to-photon latency. In one sentence, account for the difference using the supplied record. (1)

4. Recover the per-frame service demands (8 points)

For each resource, calculate service demand from the matching records:

$$D = B / C$$

where **B** is that resource's busy time during the capture and **C** is the number of completed frames. Report CPU, GPU, and SSD demand in **milliseconds per frame**. (6)

Add the three demands. Compare the sum to one field in the capture and explain why that is a useful arithmetic check. (2)

5. Find the current limit (4 points)

1. Identify the bottleneck from the three service demands and justify the choice. (1)
2. Calculate the system's maximum throughput, or ceiling, using $X_{\max} = 1000 \text{ ms/s} / D_{\max}$. Include units. (2)
3. Explain why the observed frame rate may be below this ceiling even though the bottleneck has been identified. Use the bottleneck's busy percentage, not a guess about the hardware. (1)

6. Audit the three quotes (6 points)

For quotes A, B, and C, halve only the named resource's demand. For each quote, report the new largest demand, bottleneck, and system ceiling. (3)

Choose the purchase best supported by these records, or state that none is supported. Defend the choice with the before-and-after ceiling and price. (2)

The vendor advertises a headline specification such as “twice as fast.” Explain why that statement alone does not imply twice the frame rate. (1)

7. Test the settings change and plan the re-measurement (6 points)

Your packet includes one no-purchase settings change and a supplied estimate that it halves one named stage's demand.

1. Calculate the new three demands, bottleneck, and system ceiling. Name the next bottleneck. (3)
2. Video-memory use is 94%. Explain why that number is worth watching but does not, by itself, establish the current bottleneck. (1)
3. After making the change, name **two values from the supplied records** you would collect again. For each, state the direction or value your analysis predicts. A useful verification plan could prove you wrong. (2)

8. Write the consultant's memo (4 points)

Write a short memo to the student who owns this PC. Your response must address each of the following:

- what performance was observed;
- which part currently limits it and the numeric evidence;
- what to change and the predicted ceiling;
- what not to buy, with a numeric reason; and
- what to collect in the next matched benchmark to verify the recommendation.

The memo must agree with your calculations and distinguish an observed result from a prediction.

Submit

Submit one readable PDF containing your packet identifier, answers 1–8, all calculation work, and units.